

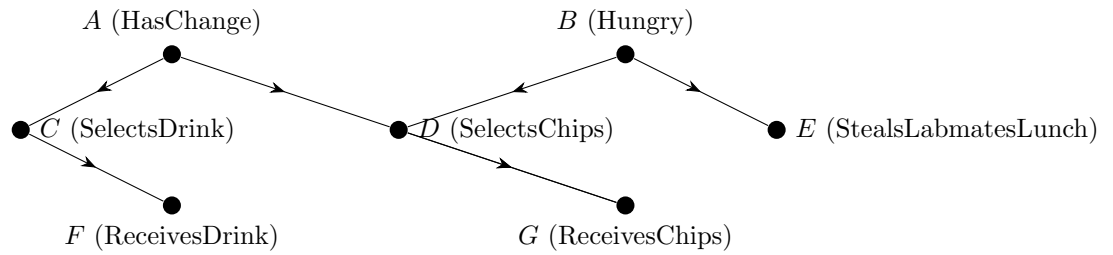
Homework 5

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Exercise 1

Consider the Bayesian network with the following DAG over Boolean variables:



with the following probability tables:

$P(A = 0)$	$P(A = 1)$	$P(B = 0)$	$P(B = 1)$
7/10	3/10	9/10	1/10

a	$P(C = 0 A = a)$	$P(C = 1 A = a)$
0	1	0
1	7/10	3/10

a	b	$P(D = 0 A = a, B = b)$	$P(D = 1 A = a, B = b)$
0	0	1	0
0	1	1	0
1	0	1/5	4/5
1	1	9/10	1/10

b	$P(E = 0 B = b)$	$P(E = 1 B = b)$
0	999999/1000000	1/1000000
1	19/20	1/20

c	$P(F = 0 C = c)$	$P(F = 1 C = c)$
0	1	0
1	1/10	9/10

d	$P(G = 0 D = d)$	$P(G = 1 D = d)$
0	1	0
1	1/10	9/10

Then, applying the Markov condition:

$$\begin{aligned}
 P(G = 1|D = 1, E = 1) &= P(G = 1|D = 1) \\
 &= 9/10
 \end{aligned}$$

Storing the probability function P in terms of its joint distribution over all seven variables would involve keeping track of probabilities for $2^7 = 128$ possible combinations of values for the variables. Since the probabilities must sum to 1, it is sufficient to keep track of $2^7 - 1 = 127$ real numbers. However, to specify the above Bayesian network, it is sufficient to keep track of only $2 \cdot (2 \cdot 2^0 + 4 \cdot 2^1 + 1 \cdot 2^2) = 28$ possible combinations of values for the variables. Since the probabilities must sum to 1, it is further sufficient to keep track of $2 \cdot 2^1 + 4 \cdot 2^1 + 1 \cdot 2^2 = 14$ real numbers.

Exercise 2

Considering the Bayesian network from Problem 1, we seek to simplify:

$$P(E = 1, D = 1, B = 1, A = 1)$$

Applying the chain rule:

$$P(E = 1, |D = 1, B = 1, A = 1)P(D = 1, B = 1, A = 1)$$

Applying the chain rule:

$$P(E = 1, |D = 1, B = 1, A = 1)P(D = 1|B = 1, A = 1)P(B = 1, A = 1)$$

Applying the chain rule:

$$P(E = 1, |D = 1, B = 1, A = 1)P(D = 1|B = 1, A = 1)P(B = 1|A = 1)P(A = 1)$$

Applying the Markov condition (since B is not a descendant of A):

$$P(E = 1, |D = 1, B = 1, A = 1)P(D = 1|B = 1, A = 1)P(B = 1)P(A = 1)$$

Applying the Markov condition (since E is not a descendant of A):

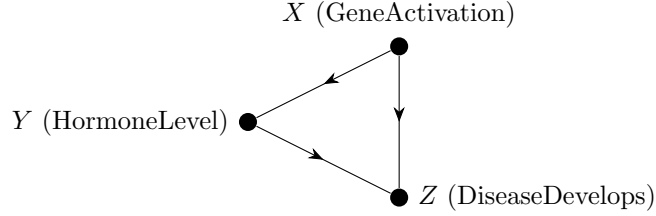
$$P(E = 1, |D = 1, B = 1)P(D = 1|B = 1, A = 1)P(B = 1)P(A = 1)$$

Applying the Markov condition (since E is not a descendant of D):

$$P(E = 1, |B = 1)P(D = 1|B = 1, A = 1)P(B = 1)P(A = 1)$$

Exercise 3

Consider the Bayesian network with the following DAG over Boolean variables:



with the following probability tables:

$P(X = 0)$	$P(X = 1)$
1/2	1/2

x	$P(Y = 0 X = x)$	$P(Y = 1 X = x)$
0	1/5	4/5
1	4/5	1/5

x	y	$P(Z = 0 X = x, Y = y)$	$P(Z = 1 X = x, Y = y)$
0	0	1	0
0	1	7/8	1/8
1	0	1/4	3/4
1	1	1/2	1/2

Then, note that by the factorization rule:

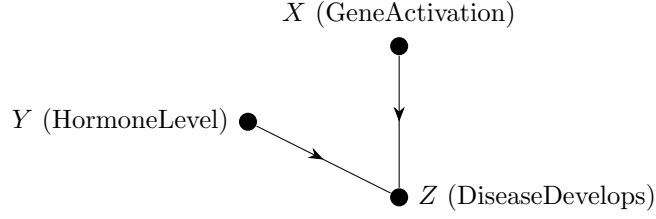
$$\begin{aligned}
 P(X = x, Y = y, Z = z) &= P(Z = z|X = x, Y = y)P(X = x, Y = y) \\
 &= P(Z = z|X = x, Y = y)P(Y = y|X = x)P(X = x)
 \end{aligned}$$

Using this factorization:

$$\begin{aligned}
 P(X = 0, Y = 0, Z = 0) &= 1 \cdot 1/5 \cdot 1/2 = 1/10 \\
 P(X = 0, Y = 0, Z = 1) &= 0 \cdot 1/5 \cdot 1/2 = 0 \\
 P(X = 0, Y = 1, Z = 0) &= 1/8 \cdot 4/5 \cdot 1/2 = 1/20 \\
 P(X = 0, Y = 1, Z = 1) &= 7/8 \cdot 4/5 \cdot 1/2 = 7/20 \\
 P(X = 1, Y = 0, Z = 0) &= 1/4 \cdot 4/5 \cdot 1/2 = 1/10 \\
 P(X = 1, Y = 0, Z = 1) &= 3/4 \cdot 4/5 \cdot 1/2 = 3/10 \\
 P(X = 1, Y = 1, Z = 0) &= 1/2 \cdot 1/5 \cdot 1/2 = 1/20 \\
 P(X = 1, Y = 1, Z = 1) &= 1/2 \cdot 1/5 \cdot 1/2 = 1/20
 \end{aligned}$$

We see that $P(Z = 1) = 0 + 7/20 + 3/10 + 1/20 = 7/10$.

To find the probability that the disease develops given that the hormone level is low ($P(Z = 1|\text{do}(Y = 0))$), we modify the causal Bayesian network as follows:



with the following probability tables:

$P(X = 0)$	$P(X = 1)$
$1/2$	$1/2$

$P(Y = 0)$	$P(Y = 1)$
1	0

x	y	$P(Z = 0 X = x, Y = y)$	$P(Z = 1 X = x, Y = y)$
0	0	1	0
0	1	$7/8$	$1/8$
1	0	$1/4$	$3/4$
1	1	$1/2$	$1/2$

Then, note that by the factorization rule:

$$\begin{aligned}
 P(X = x, Y = y, Z = z) &= P(Z = z|X = x, Y = y)P(X = x, Y = y) \\
 &= P(Z = z|X = x, Y = y)P(Y = y|X = x)P(X = x) \\
 &= P(Z = z|X = x, Y = y)P(Y = y)P(X = x)
 \end{aligned}$$

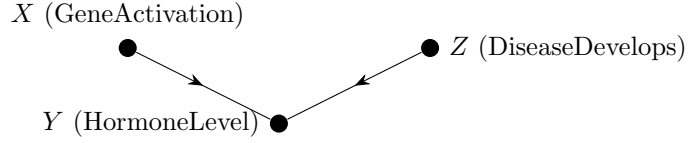
Using this factorization:

$$\begin{aligned}
 P(X = 0, Y = 0, Z = 0) &= 1 \cdot 1 \cdot 1/2 = 1/2 \\
 P(X = 0, Y = 0, Z = 1) &= 0 \cdot 1 \cdot 1/2 = 0 \\
 P(X = 0, Y = 1, Z = 0) &= 1/8 \cdot 0 \cdot 1/2 = 0 \\
 P(X = 0, Y = 1, Z = 1) &= 7/8 \cdot 0 \cdot 1/2 = 0 \\
 P(X = 1, Y = 0, Z = 0) &= 1/4 \cdot 1 \cdot 1/2 = 1/8 \\
 P(X = 1, Y = 0, Z = 1) &= 3/4 \cdot 1 \cdot 1/2 = 3/8 \\
 P(X = 1, Y = 1, Z = 0) &= 1/2 \cdot 0 \cdot 1/2 = 0 \\
 P(X = 1, Y = 1, Z = 1) &= 1/2 \cdot 0 \cdot 1/2 = 0
 \end{aligned}$$

We see that $P(Z = 1|\text{do}(Y = 0)) = 0 + 0 + 3/8 + 0 = 3/8$, and so $P(Z) > P(Z = 1|\text{do}(Y = 0))$. Lowering the hormone level is then a reasonable measure to cure the disease.

Exercise 4

Consider the Bayesian network with the following DAG over Boolean variables:



with the following probability tables:

$P(X = 0)$	$P(X = 1)$	$P(Z = 0)$	$P(Z = 1)$
1/2	1/2	3/10	7/10

x	z	$P(Y = 0 X = x, Z = z)$	$P(Y = 1 X = x, Z = z)$
0	0	2/3	1/3
0	1	0	1
1	0	2/3	1/3
1	1	6/7	1/7

Then, note that by the factorization rule:

$$\begin{aligned}
 P(X = x, Y = y, Z = z) &= P(Y = y|X = x, Z = z)P(X = x, Z = z) \\
 &= P(Y = y|X = x, Z = z)P(X = x|Z = z)P(Z = z) \\
 &= P(Y = y|X = x, Z = z)P(X = x)P(Z = z)
 \end{aligned}$$

Using this factorization:

$$\begin{aligned}
 P(X = 0, Y = 0, Z = 0) &= 2/3 \cdot 1/2 \cdot 3/10 = 1/10 \\
 P(X = 0, Y = 0, Z = 1) &= 0 \cdot 1/2 \cdot 7/10 = 0 \\
 P(X = 0, Y = 1, Z = 0) &= 1/3 \cdot 1/2 \cdot 3/10 = 1/20 \\
 P(X = 0, Y = 1, Z = 1) &= 1 \cdot 1/2 \cdot 7/10 = 7/20 \\
 P(X = 1, Y = 0, Z = 0) &= 2/3 \cdot 1/2 \cdot 3/10 = 1/10 \\
 P(X = 1, Y = 0, Z = 1) &= 6/7 \cdot 1/2 \cdot 7/10 = 3/10 \\
 P(X = 1, Y = 1, Z = 0) &= 1/3 \cdot 1/2 \cdot 3/10 = 1/20 \\
 P(X = 1, Y = 1, Z = 1) &= 1/7 \cdot 1/2 \cdot 7/10 = 1/20
 \end{aligned}$$

We see that $P(Z = 1) = 0 + 7/20 + 3/10 + 1/20 = 7/10$.

To find the probability that the disease develops given that the hormone level is low ($P(Z = 1|\text{do}(Y = 0))$), we modify the causal Bayesian network as follows:

X (GeneActivation)



Z (DiseaseDevelops)

Y (HormoneLevel)



with the following probability tables:

$P(X = 0)$	$P(X = 1)$	$P(Y = 0)$	$P(Y = 1)$	$P(Z = 0)$	$P(Z = 1)$
1/2	1/2	1	0	3/10	7/10

Then, note that by the factorization rule:

$$\begin{aligned}
 P(X = x, Y = y, Z = z) &= P(Y = y | X = x, Z = z) P(X = x, Z = z) \\
 &= P(Y = y | X = x, Z = z) P(X = x | Z = z) P(Z = z) \\
 &= P(Y = y) P(X = x) P(Z = z)
 \end{aligned}$$

Using this factorization:

$$\begin{aligned}
 P(X = 0, Y = 0, Z = 0) &= 1 \cdot 1/2 \cdot 3/10 = 3/20 \\
 P(X = 0, Y = 0, Z = 1) &= 1 \cdot 1/2 \cdot 7/10 = 7/20 \\
 P(X = 0, Y = 1, Z = 0) &= 0 \cdot 1/2 \cdot 3/10 = 0 \\
 P(X = 0, Y = 1, Z = 1) &= 0 \cdot 1/2 \cdot 7/10 = 0 \\
 P(X = 1, Y = 0, Z = 0) &= 1 \cdot 1/2 \cdot 3/10 = 3/20 \\
 P(X = 1, Y = 0, Z = 1) &= 1 \cdot 1/2 \cdot 7/10 = 7/20 \\
 P(X = 1, Y = 1, Z = 0) &= 0 \cdot 1/2 \cdot 3/10 = 0 \\
 P(X = 1, Y = 1, Z = 1) &= 0 \cdot 1/2 \cdot 7/10 = 0
 \end{aligned}$$

We see that $P(Z = 1 | \text{do}(Y = 0)) = 7/20 + 0 + 7/20 + 0 = 7/10$, and so $P(Z) = P(Z = 1 | \text{do}(Y = 0))$. Lowering the hormone level is then not a reasonable measure to cure the disease.